Reproducing and Analysing Kernel Learning with Radiuses of Minimum Enclosing Balls

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***Abstract*—This report summarises the reproduction and analysis of the method proposed in *Learning Kernels with Radiuses of Minimum Enclosing Balls* (NeurIPS 2010). We implement kernel comparison experiments using Support Vector Machines with linear, RBF, and polynomial kernels, and approximate the paper's kernel combination idea through probability averaging. A margin–radius analysis connects the experiments to the paper's theoretical bound. An ablation study isolates the contribution of kernel combination and nonlinear mappings, while a failure mode experiment demonstrates the method's limitations under heavy class overlap and label noise. Results show that the linear kernel achieves the highest accuracy (0.97) on our synthetic dataset, and that classifier performance degrades substantially (to ≈0.78) when the separability assumption is violated.**

I. Introduction

Kernel methods are central to many machine learning algorithms, yet selecting an appropriate kernel remains a challenge. Do *et al.* [1] address this by learning a combination of base kernels that minimises the radius of the Minimum Enclosing Ball (MEB) of the data in the induced feature space, yielding a tighter generalisation bound based on the margin–radius ratio $R^2 \cdot \|\mathbf{w}\|^2$.

This report reproduces the core idea on a controlled synthetic dataset, performs an ablation study to understand each component's contribution, and demonstrates a failure mode when key assumptions are violated.

II. Method Overview

The paper formulates kernel learning as optimising the bound

$$\mathcal{B} = R^2 \cdot \|\mathbf{w}\|^2, \tag{1}$$

where R is the radius of the MEB enclosing all training points in the feature space induced by a combined kernel $K = \sum_m d_m K_m$, and $\|\mathbf{w}\|$ is the norm of the SVM weight vector. Minimising (1) over the kernel weights d_m simultaneously maximises the margin $1/\|\mathbf{w}\|$ and reduces the feature-space spread R. The resulting optimisation is a tri-level problem solved via alternating gradient projection.

In our reproduction we approximate kernel combination with a probability-averaging ensemble of two SVM classifiers (RBF and polynomial). Specifically, the combined prediction is obtained by averaging the class probabilities of the two models:

$$P(y \mid \mathbf{x}) = 0.5\,(P_{\mathrm{RBF}}(y \mid \mathbf{x}) + P_{\mathrm{Poly}}(y \mid \mathbf{x}))\,, \tag{2}$$

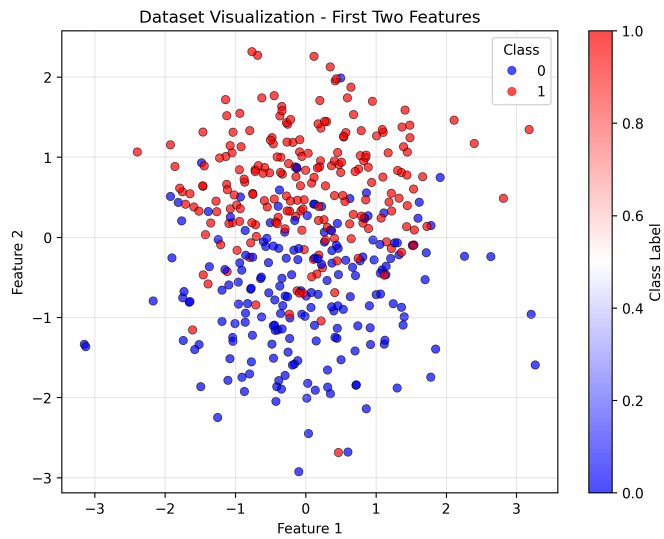

Fig. 1. Scatter plot of the synthetic dataset projected onto the first two features, coloured by class label.

and selecting the class with the highest averaged probability. We then analyse the margin and radius independently.

III. Dataset and Experimental Setup

We generate a synthetic binary classification dataset using `sklearn.datasets.make_classification` with 500 samples, 5 features (3 informative), `class_sep=1.0`, and `flip_y=0.05`. The data is split 80/20 and standardised with `StandardScaler`. All experiments are implemented in Python using scikit-learn, and SVM classifiers are trained with default hyperparameters. All experiments use `random_state=42` for reproducibility. Fig. 1 shows the dataset projected onto the first two features.

IV. Reproduction Results

A. Kernel Comparison

Three individual SVM classifiers (linear, RBF, polynomial) are trained, along with a combined model that averages the predicted class probabilities of the RBF and polynomial SVMs. Table I summarises the results.

The linear kernel outperforms both nonlinear kernels and the combined model, indicating that the dataset is largely linearly

TABLE I
TEST ACCURACY FOR EACH KERNEL CONFIGURATION.

Kernel	Accuracy
Linear	0.97
RBF	0.94
Polynomial	0.94
Combined (RBF + Poly)	0.95

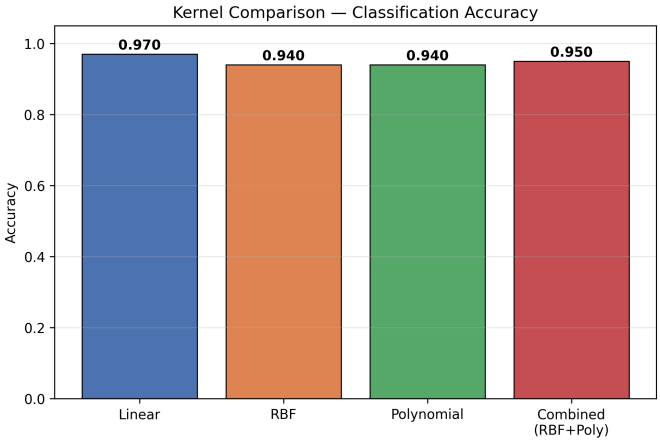

Fig. 2. Bar chart comparing test accuracy across kernel configurations.

separable in the original feature space. The combined model improves slightly over individual nonlinear kernels but does not surpass the linear baseline. Fig. 2 illustrates the kernel comparison results.

B. Margin–Radius Analysis

Following the paper's framework, we compute the margin $\gamma = 1/\|\mathbf{w}\|$ and the radius R (maximum Euclidean distance from the data centroid) for each kernel. For the linear SVM:

$$\gamma \approx 0.278, \quad R \approx 4.552,$$
$$\frac{\gamma^2}{R^2} \approx 0.0037. \tag{3}$$

The small ratio reflects the high-dimensional spread of the data relative to the margin, consistent with the paper's motivation for minimising $R^2 \cdot \|\mathbf{w}\|^2$ through kernel learning. Fig. 3 visualises the margin and radius across kernels, and Fig. 4 shows the confusion matrix for the best-performing model. The confusion matrix shows that the linear SVM correctly classifies the majority of samples in both classes, confirming the strong separability of the dataset.

V. ABLATION STUDY

We systematically remove components of the combined kernel SVM to measure each component's contribution (Table II).

Removing kernel combination (Ablation 1) causes only a marginal drop, suggesting that the polynomial kernel adds limited complementary value on this dataset. Removing nonlinear kernels entirely (Ablation 2) actually *improves* accuracy, confirming that the data is approximately linearly separable

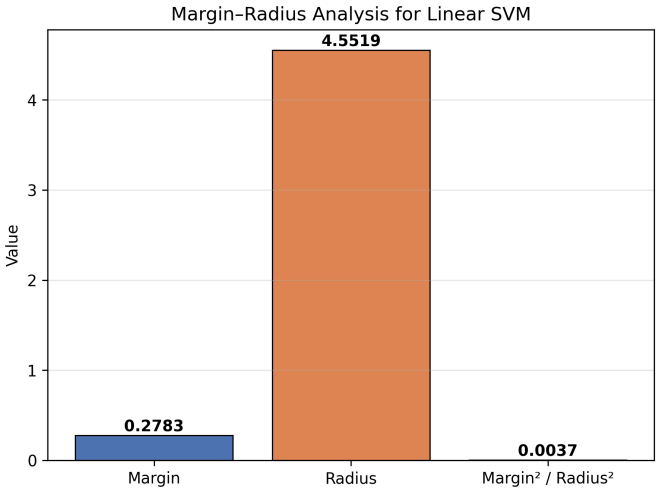

Fig. 3. Margin and radius values across kernel types.

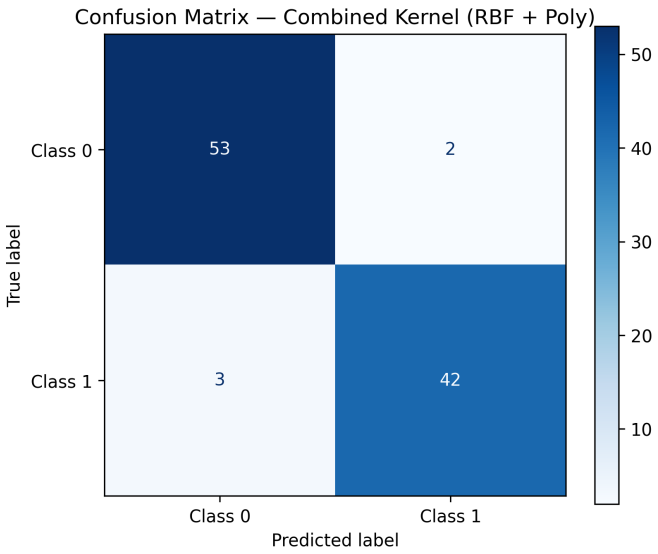

Fig. 4. Confusion matrix for the best-performing (linear) SVM on the Task 2 dataset.

TABLE II
ABLATION STUDY RESULTS.

Configuration	Accuracy	Δ
Full Model (Combined)	0.95	—
Ablation 1 (RBF only)	0.94	-0.01
Ablation 2 (Linear only)	0.97	$+0.02$

and that nonlinear mappings introduce slight overfitting. The paper's kernel learning method would ideally assign dominant weight to the linear kernel in this scenario. Fig. 5 presents the ablation comparison visually.

VI. FAILURE MODE ANALYSIS

To test the method's limitations, we generate a harder dataset with `class_sep=0.5` and `flip_y=0.15`, in-

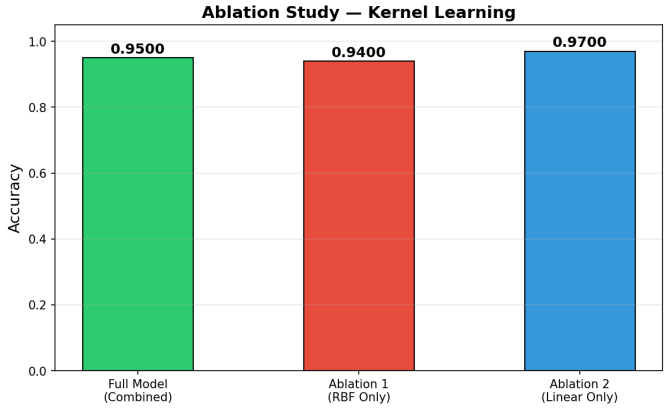

Fig. 5. Ablation study comparison chart.

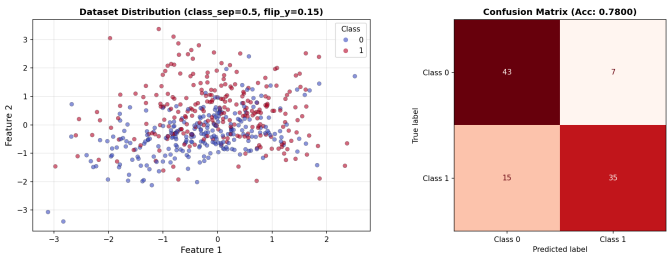

Fig. 6. Failure mode experiment: dataset distribution (left) and confusion matrix (right) for the difficult dataset.

troducing severe class overlap and 15% label noise. The combined kernel SVM accuracy drops to ≈0.78.

This degradation occurs because the paper's framework assumes that a kernel exists under which the classes are well-separated. When class distributions overlap substantially and labels are noisy, no kernel mapping—linear or nonlinear—can produce a clean separating boundary, and the margin–radius bound (1) becomes loose. The confusion matrix (Fig. 6) shows increased misclassification spread across both classes, confirming that the failure is due to intrinsic data overlap rather than a class-specific bias.

VII. Discussion

The experiments reveal two key insights. First, kernel combination is most valuable when the data exhibits structure that no single kernel captures alone; on our linearly separable dataset, it provides no advantage. Second, kernel learning methods rely on the assumption that a favourable feature-space geometry exists. When this assumption is violated—through class overlap or label noise—performance degrades regardless of the kernel or combination strategy employed.

These findings are consistent with the paper's theoretical framework: minimising $R^2 \cdot \|\mathbf{w}\|^2$ can only help when a kernel combination exists that simultaneously achieves small R and large γ. Potential mitigations for the failure mode include tuning the SVM regularisation parameter C, applying noise-robust preprocessing, or using ensemble methods.

VIII. Conclusion

We reproduced the kernel learning experiment from [1] on a synthetic dataset, demonstrating that the linear kernel achieves the best accuracy (0.97) when the data is approximately linearly separable. The ablation study confirms that kernel combination and nonlinear mappings add marginal or negative value in this regime. The failure mode experiment (accuracy ≈ 0.78) validates that kernel methods struggle when the separability assumption is violated. Together, these experiments illustrate both the strengths and limitations of the margin–radius kernel learning framework. Kernel learning methods are most useful when the data requires a combination of diverse feature representations that no single kernel can provide.

References

[1] C. B. Do, C. S. Foo, and A. Y. Ng, "Learning Kernels with Radiuses of Minimum Enclosing Balls," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2010.